GATE EE Batch 003

Human Final QA - Mobile Fillable Review Packet

Human reviewer record

Name:

Role or qualification:

Review date (YYYY-MM-DD):

Reviewer identifier (optional):

Type the attestation exactly as printed below:

This packet contains 20 checksum-bound Batch 003 General Aptitude questions covering all four official syllabus sections. Mathematical expressions are stored as canonical LaTeX and rendered through XeLaTeX.

Canonical source SHA-256: `b00075f2ae17951baccc382ea59d149e8d0df73a21fd6398bc8e2af9038c902b`

Strict Formatter report SHA-256: `e517ed2606ede6dd5f81465dc50fe1b95cbd66d9699a779c45630a2dfd10d300`

Automated prerequisite: **20 PASS / 0 REVIEW / 0 invalid**. This does not replace your human judgment.

Required procedure

1. For each item, solve the question independently before relying on the declared answer.
2. Select exactly one of PASS or FAIL for each of the five checks.
3. Select exactly one final decision: PASS, REVISE, or REJECT.
4. Explain every REVISE or REJECT decision in Notes.
5. Save this filled PDF and return it for conversion into the checksum-bound certification JSON.

Required attestation: I independently reviewed the Batch 003 questions, answers and solutions and approve only the question IDs marked PASS below.

For the qualification field, state your actual background accurately; no diploma or job title is required unless it is genuinely yours.

TMB-GATE-EE-GA-001 - revision 1

Question 1 of 20
Route: General Aptitude -> Verbal Aptitude -> Basic English Grammar
Type / marks / difficulty: MCQ / 1 / Easy

Question

For a sentence-completion task, choose the option that makes the sentence grammatically correct: “Neither the revised estimates nor the original forecast ___ consistent with the audited figures.”

Options

- A. is
- B. are
- C. were being
- D. have been

Verification reference

Declared answer: A

Independent recomputation evidence: The verb agrees with the nearer singular subject ‘forecast’, so ‘is’ is required.

Canonical solution:

With “neither … nor”, the verb agrees with the nearer subject. The nearer subject, “forecast”, is singular, so the singular verb “is” is required. Final answer: A

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-002 - revision 1

Question 2 of 20
Route: General Aptitude -> Verbal Aptitude -> Vocabulary in Context
Type / marks / difficulty: MCQ / 1 / Medium

Question

In a vocabulary-in-context review, the committee described its approval as “qualified” because two safety tests were still unresolved. Here, the word “qualified” most nearly means

Options

- A. unconditional
- B. limited by conditions
- C. professionally trained
- D. officially certified

Verification reference

Declared answer: B

Independent recomputation evidence: Unresolved conditions limit the approval; ‘qualified’ therefore means conditional.

Canonical solution:

The unresolved tests restrict the approval. In this context, “qualified” means conditional or limited rather than unconditional, trained, or formally certified. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-003 - revision 1

Question 3 of 20
Route: General Aptitude -> Verbal Aptitude -> Reading and Comprehension
Type / marks / difficulty: MCQ / 2 / Medium

Question

Read the following passage and answer the reading-comprehension question. An office tested a flexible starting-time policy for four weeks. Among the 120 employees who voluntarily submitted travel data, the recorded mean commute duration fell from 52 minutes before the pilot to 46 minutes during it. No travel data were collected from nonparticipants. Which conclusion is best supported?

Options

- A. The policy reduced the commute duration of every employee.
- B. Among the reporting employees, the recorded mean commute duration was 6 minutes lower during the pilot.
- C. The nonparticipants experienced the same mean reduction as the participants.
- D. The pilot proves that traffic volume across the entire city decreased.

Verification reference

Declared answer: B

Independent recomputation evidence: Only the reporting sample supports a $52 - 46 = 6$ minute mean reduction.

Canonical solution:

The stated measurements establish only that the mean for the reporting group changed from 52 to 46 minutes, a reduction of 6 minutes. Voluntary participation prevents conclusions about every employee, nonparticipants, or citywide traffic. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-004 - revision 1

Question 4 of 20
Route: General Aptitude -> Verbal Aptitude -> Reading and Comprehension
Type / marks / difficulty: MSQ / 2 / Hard

Question

Read the following passage and select all supported statements. A fault-prediction model scored 68% on a fixed held-out sample from Plant B. After recalibration using a separate training sample from Plant B, it scored 81% on that same held-out sample. No post-recalibration test was performed at Plant A.

Options

- A. The measured accuracy on the held-out Plant B sample increased by 13 percentage points.
- B. Recalibration guarantees improved accuracy on every future case from Plant B.
- C. The post-recalibration accuracy at Plant A is unknown from the given information.
- D. Relative to the original 68% score, the measured increase is approximately 19.1%.

Verification reference

Declared answer: A, C, D

Independent recomputation evidence: The increase is 13 percentage points and $13/68 = 19.1176\%$; Plant A was not retested.

Canonical solution:

The percentage-point increase is $81 - 68 = 13$. The relative increase is $\frac{13}{68} \times 100\% = 19.1176\%$, approximately 19.1%. No new Plant A result is supplied, and one held-out result cannot guarantee every future outcome. Final answer: A, C, D

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS☐ REVISE☐ REJECT☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-005 - revision 1

Question 5 of 20
Route: General Aptitude -> Verbal Aptitude -> Narrative Sequencing
Type / marks / difficulty: MCQ / 1 / Medium

Question

For a narrative-sequencing task, arrange the following sentences into the most coherent order. P: Only then could results from repeated tests be compared. Q: The initial instruction merely asked the team to improve reliability. R: The team therefore defined failure as more than two interruptions per 100 operating hours. S: Without a numerical criterion, however, success could not be judged.

Options

- A. Q-S-R-P
- B. Q-R-S-P
- C. S-Q-P-R
- D. R-Q-S-P

Verification reference

Declared answer: A

Independent recomputation evidence: The vague instruction Q is criticized by S, quantified by R, and made comparable by P.

Canonical solution:

Q introduces the vague instruction. S explains its defect, R supplies the resulting measurable definition, and P states what that definition then enables. The coherent order is Q-S-R-P. Final answer: A

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-006 - revision 1

Question 6 of 20
Route: General Aptitude -> Quantitative Aptitude -> Ratios and Percentages
Type / marks / difficulty: NAT / 1 / Easy

Question

In a numerical-computation problem, the listed price of an item is increased by 20% and the increased price is then reduced by 10%. Enter the net percentage increase relative to the original price.

Verification reference

Declared answer: 8

Independent recomputation evidence: Successive multipliers give $1.20 \times 0.90 = 1.08$, hence an 8% net increase.

Canonical solution:

Taking the original price as 100, the successive prices are $100 \times 1.20 = 120$ and $120 \times 0.90 = 108$. The net increase is therefore $108 - 100 = 8\%$. Final answer: 8

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-007 - revision 1

Question 7 of 20
Route: General Aptitude -> Quantitative Aptitude -> Ratios and Percentages
Type / marks / difficulty: NAT / 2 / Medium

Question

A 40-litre tank contains a uniform solution that is 25% salt by volume. A quantity x litres is removed and replaced by the same quantity of water. If the resulting solution is 16% salt, determine x in litres.

Verification reference

Declared answer: 14.4

Independent recomputation evidence: Salt balance gives $10 - 0.25x = 6.4$, so $x = 14.4$ litres.

Canonical solution:

Initially the tank contains $0.25(40) = 10$ litres of salt. Removing x litres of uniform solution removes $0.25x$ litres of salt. Replacement with water leaves the final salt volume as $10 - 0.25x$. Hence $10 - 0.25x = 0.16(40) = 6.4$, giving $x = 14.4$ litres. Final answer: 14.4

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-008 - revision 1

Question 8 of 20
Route: General Aptitude -> Quantitative Aptitude -> Data Interpretation
Type / marks / difficulty: NAT / 1 / Medium

Question

A data-interpretation table gives inspection results for three quarters.

Quarter	Accepted	Rejected
Q1	84	16
Q2	108	12
Q3	126	14

Enter the overall acceptance percentage, rounded to two decimal places.

Verification reference

Declared answer: 88.33

Independent recomputation evidence: Accepted $= 318$ and inspected $= 360$; $100(318/360) = 88.33\%$ after rounding.

Canonical solution:

The total accepted count is $84+108+126 = 318$, while the total inspected count is $318+(16+12+14) = 360$. Thus the overall acceptance percentage is $\frac{318}{360} \times 100\% = 88.333\ldots\%$, which rounds to 88.33%. Final answer: 88.33

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-009 - revision 1

Question 9 of 20
Route: General Aptitude -> Quantitative Aptitude -> Powers, Exponents and Logarithms
Type / marks / difficulty: NAT / 2 / Hard

Question

Let $x > 1$ satisfy $\log_2 x + \log_x 2 = \frac{5}{2}$. If all possible values of x are included, enter their sum rounded to three decimal places.

Verification reference

Declared answer: 5.414

Independent recomputation evidence: For $y = \log_2(x)$, $y + 1/y = 5/2$ gives $y = 2$ or $1/2$; the x values sum to $4 + \sqrt{2} = 5.414$.

Canonical solution:

Set $y = \log_2 x$, so $y > 0$ and $\log_x 2 = \frac{1}{y}$. Then $y + \frac{1}{y} = \frac{5}{2}$, or $2y^2 - 5y + 2 = 0$. Hence $y = 2$ or $y = \frac{1}{2}$, giving $x = 4$ or $x = \sqrt{2}$. Their sum is $4 + \sqrt{2} = 5.414213\ldots$, which rounds to 5.414. Final answer: 5.414

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-010 - revision 1

Question 10 of 20
Route: General Aptitude -> Quantitative Aptitude -> Permutations and Combinations
Type / marks / difficulty: MCQ / 2 / Medium

Question

In a permutations-and-combinations problem, six distinct jobs A, B, C, D, E and F are arranged in a sequence. Job A must occur before job B, and jobs C and D must not be adjacent. The number of valid sequences is

Options

- A. 120
- B. 180
- C. 240
- D. 360

Verification reference

Declared answer: C

Independent recomputation evidence: There are 360 orders with A before B; 120 also have C and D adjacent, leaving 240.

Canonical solution:

Exactly half of the $6! = 720$ sequences have A before B, giving 360. If C and D are adjacent, treating them as one block gives $2(5!) = 240$ sequences; by symmetry, half of these, namely 120, have A before B. Therefore the valid count is $360 - 120 = 240$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS☐ REVISE☐ REJECT☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-011 - revision 1

Question 11 of 20
Route: General Aptitude -> Quantitative Aptitude -> Mensuration and Geometry
Type / marks / difficulty: NAT / 1 / Easy

Question

In a mensuration problem, the area of a square is increased by 21%. Enter the percentage increase in its side length.

Verification reference

Declared answer: 10

Independent recomputation evidence: The side multiplier is $\sqrt{1.21} = 1.1$, hence a 10% increase.

Canonical solution:

If the original side is s, the new side is $s\sqrt{1.21} = 1.1s$. Hence the side length increases by 10%. Final answer: 10

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-012 - revision 1

Question 12 of 20
Route: General Aptitude -> Quantitative Aptitude -> Elementary Statistics and Probability
Type / marks / difficulty: NAT / 2 / Hard

Question

Three components fail independently during a test with probabilities 0.1, 0.2 and 0.3, respectively. The system fails if at least two components fail. Enter the probability of system failure.

Verification reference

Declared answer: 0.098

Independent recomputation evidence: Exactly two failures total $0.014 + 0.024 + 0.054$; all three add 0.006, giving 0.098.

Canonical solution:

The probabilities for exactly two failures are $(0.1)(0.2)(0.7) = 0.014$, $(0.1)(0.8)(0.3) = 0.024$, and $(0.9)(0.2)(0.3) = 0.054$. The probability that all three fail is $(0.1)(0.2)(0.3) = 0.006$. Therefore the required probability is $0.014 + 0.024 + 0.054 + 0.006 = 0.098$. Final answer: 0.098

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-013 - revision 1

Question 13 of 20
Route: General Aptitude -> Analytical Aptitude -> Logic: Deduction and Induction
Type / marks / difficulty: MSQ / 1 / Medium

Question

For a logic-deduction task, consider the premises: (i) every relay trip is logged; (ii) some logged events are manually generated; and (iii) no manually generated event is a fault. Select all conclusions that must follow.

Options

- A. No manually generated event is a fault.
- B. Some logged events are not faults.
- C. Some relay trips are not faults.
- D. Every logged event is a relay trip.

Verification reference

Declared answer: A, B

Independent recomputation evidence: The premises state A; an existing logged manual event is a non-fault, establishing B.

Canonical solution:

Statement A repeats premise (iii). By premise (ii), at least one logged event is manually generated; premise (iii) makes that event a non-fault, so B follows. Nothing states that any relay trip is a non-fault, and premise (i) does not justify its converse. Final answer: A, B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-014 - revision 1

Question 14 of 20
Route: General Aptitude -> Analytical Aptitude -> Logic: Deduction and Induction
Type / marks / difficulty: MCQ / 2 / Medium

Question

In an analytical-aptitude scheduling problem, four inspections P, Q, R and S occupy positions 1 through 4, one per position. P is before Q, R is immediately after P, and S is not fourth. Which statement must be true?

Options

- A. P is first.
- B. Q is fourth.
- C. R is second.
- D. S is first.

Verification reference

Declared answer: B

Independent recomputation evidence: The only feasible orders are P-R-S-Q and S-P-R-Q; Q is fourth in both.

Canonical solution:

If P is first, R is second, and the condition on S forces the order P-R-S-Q. If P is second, R is third, forcing S-P-R-Q. P cannot be third because Q would then have no later position. Both feasible orders place Q fourth, while none of the other statements holds in both. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-015 - revision 1

Question 15 of 20
Route: General Aptitude -> Analytical Aptitude -> Analogy and Numerical Relations
Type / marks / difficulty: MCQ / 1 / Easy

Question

In a numerical-analogy question, the pairs $3 : 15$, $5 : 35$ and $8 : 80$ follow the same rule. Under that rule, 11 is paired with

Options

- A. 121
- B. 132
- C. 143
- D. 154

Verification reference

Declared answer: C

Independent recomputation evidence: The common rule is $n(n + 2)$, and $11 \times 13 = 143$.

Canonical solution:

Each second number is $n(n + 2)$: $3(5) = 15$, $5(7) = 35$, and $8(10) = 80$. Thus 11 is paired with $11(13) = 143$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-016 - revision 1

Question 16 of 20
Route: General Aptitude -> Analytical Aptitude -> Logic: Deduction and Induction
Type / marks / difficulty: MSQ / 2 / Hard

Question

For a logic-deduction task, use these premises: (i) every device that passed test T was calibrated; (ii) some calibrated devices failed test S; and (iii) no device that failed test S was shipped. Select all conclusions that necessarily follow.

Options

- A. Some calibrated devices were not shipped.
- B. Every shipped device passed test T.
- C. Some devices that passed test T were shipped.
- D. No uncalibrated device passed test T.

Verification reference

Declared answer: A, D

Independent recomputation evidence: A follows from the existing calibrated S-failure; D is the contrapositive of passed-T implies calibrated.

Canonical solution:

The devices asserted in premise (ii) are calibrated and failed S; premise (iii) makes them not shipped, so A follows. Premise (i) is equivalent to saying that no uncalibrated device passed T, so D follows. The premises do not establish B or the existence claim in C. Final answer: A, D

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-017 - revision 1

Question 17 of 20
Route: General Aptitude -> Spatial Aptitude -> Transformation of Shapes
Type / marks / difficulty: MCQ / 1 / Easy

Question

In a spatial-aptitude transformation, an arrow initially points north. It is rotated 90° clockwise and then mirrored across an east-west line. Its final direction is

Options

- A. east
- B. west
- C. north
- D. south

Verification reference

Declared answer: A

Independent recomputation evidence: North rotated 90° clockwise becomes east; an east-west mirror leaves east unchanged.

Canonical solution:

The clockwise rotation changes north to east. Reflection across an east-west line reverses the north-south component but leaves an east-pointing arrow unchanged. The final direction is east. Final answer: A

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-018 - revision 1

Question 18 of 20
Route: General Aptitude -> Spatial Aptitude -> Paper Folding and Cutting
Type / marks / difficulty: NAT / 2 / Medium

Question

For a paper-folding problem, a square sheet is folded once along its vertical midline and then once along its horizontal midline. In the final folded quarter, hole P is punched strictly inside and away from both creases. Hole Q is punched on the horizontal crease created by the second fold but away from the vertical crease and the original sheet boundary. After complete unfolding, enter the total number of distinct holes.

Verification reference

Declared answer: 6

Independent recomputation evidence: The interior punch produces four holes; the punch on one crease produces two, for six total.

Canonical solution:

Hole P is away from both creases, so each unfolding doubles it and produces $2 \times 2 = 4$ holes. Hole Q lies on the second-fold crease, so the horizontal unfolding does not duplicate it; the vertical unfolding does, producing 2 holes. The locations are distinct, so the total is $4 + 2 = 6$. Final answer: 6

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-019 - revision 1

Question 19 of 20
Route: General Aptitude -> Spatial Aptitude -> Three-Dimensional Grouping
Type / marks / difficulty: MSQ / 1 / Medium

Question

In a three-dimensional spatial-aptitude problem, a cube has distinct face labels P, Q, R, S, T and U. The opposite face pairs are P-U, Q-T and R-S. Select all triples that can meet at a single vertex.

Options

- A. P, Q, R
- B. P, U, T
- C. Q, R, S
- D. P, T, S

Verification reference

Declared answer: A, D

Independent recomputation evidence: A cube vertex contains one face from each opposite pair; only A and D satisfy that rule.

Canonical solution:

A vertex contains exactly one face from each opposite pair. P-Q-R and P-T-S each select one label from P-U, one from Q-T, and one from R-S, so they are possible. P-U-T contains the opposite pair P-U, and Q-R-S contains the opposite pair R-S. Final answer: A, D

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-GA-020 - revision 1

Question 20 of 20
Route: General Aptitude -> Spatial Aptitude -> Three-Dimensional Patterns
Type / marks / difficulty: NAT / 2 / Hard

Question

In a three-dimensional pattern problem, every outer face of a solid cube is painted and the cube is divided into $5 \times 5 \times 5$ congruent smaller cubes. All smaller cubes having exactly one painted face are removed. Enter the number of smaller cubes that remain.

Verification reference

Declared answer: 71

Independent recomputation evidence: Exactly-one-painted-face cubes number $6(5-2)^2 = 54$; $125 - 54 = 71$ remain.

Canonical solution:

On each large face, the smaller cubes with exactly one painted face form the interior 3×3 array, so their number is $6(3^2) = 54$. The original cube contains $5^3 = 125$ smaller cubes. Hence $125 - 54 = 71$ remain. Final answer: 71

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

Review completion

Complete this page only after reviewing all 20 questions.

Final reviewed-results decision: APPROVE REVIEWED RESULTS ☐

Reviewer typed signature:

Final comments:

Paper eligibility remains blocked until the completed human record is validated and only explicitly approved IDs are promoted.